

Performance Evaluation of Mutual Funds: The Issue of Reporting and Disclosure

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Abstract

The mutual fund industry spanning over almost two decades now, has seen its share of success and failure. It has evolved dynamically and grown substantially. The need for regulation and transparency assumes greater importance in such a growing industry given the background of complex financial environment. The reporting and disclosure requirements with regard to financial performance of mutual fund schemes is a key issue to ensure informed investment decisions by investors. In this study, we explore such norms for mutual funds in India and suggest specific risk-adjusted performance evaluation measures to be mandatorily reported by mutual fund houses for each of their schemes based on the degree of rank correlation among such measures. Our methodology consists of survey of the existing regulation and calculation of Spearman's and Kendall's rank correlation coefficient to identify appropriate measures of risk adjusted performance. We conclude that there is a severe lack of risk adjusted performance reporting and disclosure requirements in India, coupled with inadequate clarity on measurement of risk and assessment of performance. We suggest half-yearly reports of mutual funds should include one risk-adjusted measure from each of the four different classes of risk measures discussed in the paper to ensure informed investment decisions.

Key Words: *Risk-Adjusted Performance Measures, Rank Correlation, Value at Risk, Jensen's Alpha, Modified Cornish-Fisher VaR*

JEL Classification Codes: D81; G10; G11; G20

1. Introduction

The Indian mutual fund industry which was once synonymous with Unit Trust of India (UTI), set up by Government of India in 1963, was opened up to the entry of private players a little over twenty years ago. It is an achievement that we have reached a mark of almost INR 10 trillion of assets under management as of May 2014¹. It has been a little over 20 years since the asset management industry was opened up to the entry of new players. The objective was to expand the business by widening and deepening the market for asset management products

Considering the industry is still relatively young, its evolution and growth over the two decades is impressive. This is true in all aspects and not just about the scale of growth of assets under management (AuM). The regulatory regime kept pace with the changing environment and the AuM of the asset management industry grew from INR 470 billion in 1993 to INR 1396 billion in 2004 and to INR 8252 billion in 2014¹. It has been two decades since the regulators set the industry on the path to growth and expansion. In order to help chart the course for the next decade, this is perhaps a good time to step back to take a look and assess the current reporting regulations of mutual funds.

With SEBI as the regulatory authority for mutual fund schemes in India, strict guidelines are in force with regard to registration, operations, reporting and disclosure requirements of a mutual fund scheme. In this study, we aim to look at the aspect of disclosure and reporting of the performance of a mutual fund scheme. This is an important issue for the investor keeping in view the plethora of schemes in the market and low financial awareness of retail investors. We explore the current guidelines of disclosure requirements with regard to performance evaluation of schemes in India. We also conduct a ranking comparison of 80 equity schemes in India spread across eleven asset management companies on the basis of ten performance evaluation ratios based on different definitions of risk. The idea is to find out if the rankings of these 80 schemes remain consistent across the ten ratios. This can provide us with an indication as to which ratios are relevant for reporting to compare the performance of different mutual fund schemes in the market.

¹CII Report with PwC “Indian mutual fund industry Challenging the status quo, setting the growth path”, 2014

The paper is structured as follows. In the next section we discuss the need and relevance of the study. In the third section we discuss the objectives of the study. In the fourth section we describe the data and methodology used in the study. Fifth section constitutes the analysis and key findings of the study. Sixth section discusses the implications of the study.

2. Need of the Study

Although mutual funds have come up as an attractive investment avenue for the investors as they are managed by experts having keen insights into the capital markets, they also carry certain risks. An evaluation of the performance of mutual funds is imperative for investors wishing to avoid additional risks on their investments. The main concern of an investor is to take a decision regarding whether or not to invest in a particular mutual fund. Factors considered by investors in order to help them with decision-making often include fund ranking and fund performance.

The history of portfolio performance starts back from the Markowitz portfolio theory introduced in 1959, which explained the formation/ construction of the portfolio to maximize the return for same level risk or to minimize the risk for same level of return. This theory contributed to the development of asset pricing models (focusing on the relationship between the price and the risk). The development of Capital Asset Pricing Model (CAPM) that defines the returns taking into consideration the non-diversified risk i.e. the excess returns against funds beta (β) (Treyner, 1965). (Sharpe, 1966) suggested to measure the risk against the total risk instead of only fund's β i.e. return on market portfolio. Later Jensen (1968) proposed alpha (α) as the measure of risk, which takes into account the differences between the actual and expected return which in other words is the abnormal returns of the security.

Sortino and Price (1994) replace the standard deviation in the Sharpe's measure with downside deviation. In 1997, (Modigliani and Modigliani, 1997) modified the Sharpe's measure by measuring the M^2 ratio which multiplies the Sharpe's measure by the benchmark standard deviation and add it to the risk-free rate of return. Then, Pedersen and Rudholm (2003) propose the information ratio (IR) which has advantages over Sharpe ratio as "it adjust the returns of a fund to the risk levels in an unmanaged stock market index and then evaluate the returns on the risk-matched fund. Second, it reports the performance of a portfolio as a percentage easily understood by a lay investor" (Arugaslan et al., 2008).

There are number of measures available so in the paper we try to evaluate these measures by ranking our selected schemes based on these measures so as to find out which measures give the similar or contrasting ranking of funds. (Grau-Carles, Sainz, Otamendi, & Doncel, 2009) consider performance measures that take VaR into account and compare the rating of the funds under the Basel II risk management framework. Treynor measure according to (Hsieh & Hodnett, 2013) is more suitable for a broader portfolio, while the information ratio is useful for evaluating hedge funds. The Sharpe ratio and M-squared turn up to the same results(Nagar, Gupta, & Krishnamurthi, 2013). (Simons, 1998) checks the reliability and accuracy of the performance measures techniques, provides evidences to choose techniques as per the requirements of the investors.

In its guidelines, SEBI has mandated mutual funds to disclose the performance of their schemes for the previous six months, one year, three years, five years and since the date of launch, while publishing their half-yearly results. Mutual funds are also required to disclose in the offer document as well as the half-yearly results the performance of the appropriate market index along with the performance of the schemes. Depending on the investment objective of the mutual funds, SEBI has also indicated the appropriate benchmark indices for various kinds of schemes. Further, the performance of the mutual funds is required to be reviewed by the trustees.

Despite the above mentioned performance disclosure requirements, there is a need to revisit the issue from the perspective of reporting ‘risk-adjusted performance’ rather than just ‘performance’. The superiority of any investment alternative with respect to other investible assets is appropriate only if the returns per unit of risk are compared. Comparing only returns is certainly not a correct approach.

The existing guidelines do not mention any mandatory requirements for reporting and disclosure of risk adjusted performance of mutual funds schemes in India. This issue assumes even higher importance due to the existence of multiple definitions of risk by academicians and researchers. The existence of different risk measures which vary in terms of their outlook towards the phenomenon of risk and its statistical quantification thereof is evident of the fact how a concept can be viewed differently by different people. We take up this issue of specification of the risk-adjusted measures fit for reporting and disclosure requirements based on whether and which of

these measures give congruent or contrasting rankings of 80 mutual funds schemes selected for the study. It is ideally rational to have reporting requirements for those risk-adjusted performance measures which give contrasting rankings so that the investor can choose to make his investment decisions based on his own interpretation of risk.

3. Objectives of the Study

Keeping in view the growing mutual fund industry in India and rising financial awareness and literacy levels, we explore the existing regulatory reporting and disclosure requirements as mandated by SEBI with regard to performance of mutual fund schemes. We also inspect the existing regulatory reporting and disclosure requirements as mandated by SEBI with regard to risk-adjusted performance of mutual fund schemes. Finally, we analyze whether different methods of risk measurement yield different ranking of equity portfolios on a risk adjusted basis. This idea has been applied to ten risk adjusted measures spread across four different groups based on how they define and measure risk. These groups are volatility based measures, sensitivity based measures, downside-risk based measures and tail based measures. We hypothesize that there is no pair-wise rank correlation between any of the aforementioned groups of measures.

4. Data and Methodology

To serve our first two objectives the key guiding document is the SEBI (MUTUAL FUNDS) REGULATIONS, 1996 and related amendments thereof. The other source of the review is the website of AMFI, a nodal association of mutual funds across India. AMFI is an industry standards organization in India in the mutual funds sector.

For the third objective, we took the monthly Net Asset Value (NAV) data for equity schemes of those fund houses which had such data available for at least four or more equity schemes over a period of 10 years starting 30th July 2005 till 30th June 2015. In this way we shortlisted 80 equity schemes spread across 10 asset management companies operating in India. The monthly NAV data for all schemes has been taken from ACE mutual funds database. We retrieved the corresponding monthly data for Government of India 91-day Treasury Bill Yield and BSE 500 values from Bloomberg database. The Treasury bill yields were annual yields and were thereby converted into monthly yields using Effective Interest Rate formula and used as a proxy for risk

free rate. Since our analysis focuses only on equity growth mutual funds which are diversified portfolios, BSE 500 has been taken as the benchmark index to evaluate mutual fund performance. We converted the monthly NAV and BSE 500 values into monthly returns by taking the log first difference. The final sample consisted of 80 mutual funds having data for 120 months.

The descriptive statistics of all the eighty mutual funds under consideration for the study have been provided in the Table 2 in the Appendix section. The mean monthly log return across 80 schemes selected for the study is 1.21% whereas the corresponding average standard deviation for these monthly returns is 7.32%. In summary all 80 schemes have a leptokurtic and negatively skewed (except four schemes) monthly return distribution. Hence we can observe that the null hypothesis of normality in Jarque-Bera Test is rejected in all schemes even at 95% confidence level. This means that all 80 return distributions in our study are not normal.

In the present study four categories of performance measures are used: Volatility based, Sensitivity based, Downside-risk based, and Tail based measures. Each of these categories incorporates different measures.

4.1 Volatility Based Measures:

4.1.1 Sharpe Ratio – Sharpe ratio is used to measure performance when returns are normally distributed. Also known as reward-to-variability ratio, Sharpe ratio is calculated as follows:

$$\text{Sharpe Ratio} = \frac{\text{Fund's excess return}}{\text{Standard deviation of fund}}$$

where excess return represents the return over and above the risk-free rate

4.1.2 M² Ratio – Also known as Modigliani measure, this ratio expresses a fund's performance relative to the market in percentage terms.

$$\text{Modigliani Measure} = \frac{\text{Fund's excess return} \times \text{Standard deviation of benchmark index excess return}}{\text{Standard deviation of fund's excess return}} + \text{Risk-free rate of return}$$

4.2 Sensitivity Based Measures:

- 4.2.1 Treynor's Ratio – Treynor's ratio also measures the association between portfolio returns in excess of risk-free rate of return divided by risk, where, risk is defined as beta or sensitivity of portfolio returns to returns of a benchmark index.

$$\text{Treynor's Ratio} = \frac{\text{Fund's excess return}}{\text{Beta}}$$

- 4.2.2 Jensen Alpha – Jensen's alpha provides a long-term view of fund manager's ability to add value or provide superior fund returns. This measure indicates the difference between a portfolio's actual and expected return given its level of systematic risk. Jensen's Alpha is computed using the following regression equation:

$$R_i - R_f = \alpha + \beta [R_m - R_f] + \varepsilon_i$$

where R_i is the return of the fund, R_f is the risk-free rate of return, α is the Jensen's Alpha, β is the Beta or sensitivity, R_m is the return of benchmark index, and ε is the error term

- 4.2.3 Information Ratio – It is derived by comparing a fund to its benchmark. Information ratio measures the ability of a fund manager to generate returns over and above a benchmark after adjusting it for tracking error. Tracking error is defined as standard deviation of difference between fund return and benchmark index return.

$$\text{Information Ratio} = \frac{(\text{Fund return} - \text{Benchmark index return})}{\text{Standard deviation (Fund return} - \text{Benchmark index return)}}$$

- 4.3 Downside Risk Based Measures:

- 4.3.1 Sortino Ratio – In Sharpe ratio, standard deviation in the denominator does not differentiate between good and bad deviations, however, most investors being risk averse dislike losses but are happy with excess returns. Sortino ratio utilizes only returns earned below a minimum acceptable limit to be used to get a true picture of the fund's performance.

$$\text{Sortino Ratio} = \frac{\text{Fund's excess return}}{\text{Semi-standard Deviation}}$$

4.4 Tail Based Measures

- 4.4.1 Historical Simulation Value at Risk - Historical Simulation is the most widely implemented Non-parametric approach for estimating Value at Risk (VaR). This method uses the empirical distribution of financial returns, thus $\text{VaR}(\alpha)$ is the α quantile of empirical distribution. We have calculated a Modified Sharpe ratio by dividing the excess returns of the scheme over and above the risk-free rate of return by Historical Simulation VaR. We call this measure Modified Sharpe Historical VaR.
- 4.4.2 Normal Value at Risk - Standard VaR estimates the potential loss over a specific time horizon for a given probability. It is a forward-looking measure but it is critiqued for being ignorant about the extreme loss a portfolio could incur in the left tail of the distribution. We have calculated a Modified Sharpe ratio by dividing the excess returns of the scheme over and above the risk-free rate of return by NormalVaR. We call this measure Modified Sharpe NormalVaR.
- 4.4.3 Modified Cornish Fisher Value at Risk - When returns do not follow normal distribution, Normal VaR is not efficient and to tackle the issue of non-normal distribution, Modified VaR (MVaR) is used. This MVaR is based on Cornish-Fisher expansion (Cornish and Fisher, 1937) to estimate VaR in the left-tail of the distribution. Cavenaile and Lejeune (2012) explain that MVaR should not be used with confidence levels below 95.84% so as to maintain consistency with investors' preferences for kurtosis. Thus, estimation of MVaR with 95% confidence level may give inconsistent results. As a result, we compute our results using 99% confidence level for MVaR, hence for all the other measures the confidence level is also same except the expected shortfall for which 95% confidence level is maintained. We have calculated a Modified Sharpe ratio by dividing the excess returns of the scheme over and above the risk-free rate of return by Cornish Fischer VaR. We call this measure Modified Sharpe Cornish-Fischer VaR.
- 4.4.4 Expected Shortfall - While the VaR tells us nothing more than to expect a loss higher than the VaR itself, Expected Shortfall measures the expected value of our losses if we

get a loss in excess of VaR. So that Expected Shortfall measure tells us what to expect in a bad state. We have calculated a Modified Sharpe ratio by dividing the excess returns of the scheme over and above the risk-free rate of return by Expected Shortfall. We call this measure Modified Sharpe Expected Shortfall VaR.

Further, ranking of the mutual fund schemes are done according to the nine different measures used in the study. The correlation among the ranks of the schemes are calculated using following two rank correlation measures.

4.5 Spearman rank Correlation coefficient- It is a non-parametric measure of statistical dependence of variables. The formula used In test statistic is as follows:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where, $d = x_i - y_i$ is the difference between ranks, n = total number of observations, ρ = Spearman correlation coefficient

4.6 Kendall τ - Kendall rank correlation coefficient is used to measure association between two or more than two quantiles. The tau- test is the non-parametric hypothesis testing for analyzing statistical dependence, used the Kendall's τ methodology.

The Kendal τ statistic is defined as follows:

$$\tau = (no. of concordant pairs) - (no. of discordant pairs) / (\frac{1}{2}n(n - 1))$$

Where, any pair of observations (x_i, y_i) and (x_j, y_j) are said to be concordant if the ranks for both elements agree: that is, if both $x_i > x_j$ and $y_i > y_j$ or if both $x_i < x_j$ and $y_i < y_j$.

5. Analysis and Interpretation

We explore the existing reporting and disclosure requirements as prescribed by SEBI with regard to performance of mutual fund schemes in India through the updated and amended SEBI (Mutual Funds) Regulations² which were first introduced in 1996. We observe that most directions and guidelines relate to reporting of statements of accounts including the balance sheet and the profit

² SEBI (Mutual Funds) Regulations, 1996 from www.sebi.gov.in

and loss account for the fund and in respect of each scheme on a periodic basis. The composition of the fund portfolio is also mandated to be disclosed by SEBI on a periodic basis.

Other than that there is ambiguity on reporting of performance of a mutual fund scheme. There is lack of clarity on how is this performance reporting to be quantifies in terms of numbers. The definition of risk is not specified. Although the guidelines do prescribe the categorization of schemes on the basis of risk in three zones but it does not specify the method of measurement of these risks. There is no mention of reporting the risk adjusted performance of a scheme at all. No mathematical methods are specified in order to track performance of a scheme. Therefore we can clearly observe the absence of risk adjusted performance reporting and disclosures as mandated by SEBI for mutual fund schemes in India.

With regard to AMFI, all guidelines are suggestive in nature. There are no hard and clear requirements with respect to reporting the performance of a mutual fund scheme in India. The guidelines do mention the disclosure of performance reports to the investors but the content for preparation of such reports is not specified. However, maintenance of consistency in the performance evaluation method is prescribed. Member fund houses are restrained from making any exaggerated claims regarding performance of their schemes but the concept of performance or risk is not discussed. Similar to the case of SEBI, there is no mention of reporting the risk adjusted performance of a scheme at all. To summarize, there is excessive lack of mandatory reporting and disclosure requirements with regard to performance of a mutual fund scheme in India in general and risk adjusted performance in general.

We now discuss the results of the results of the third and final objective of comparing the ranks obtained by 80 schemes according to 10 different risk-adjusted performance measures spread across four classes of risk measurement i.e. Volatility based, Sensitivity based, Downside-risk based, and Tail based measures.

Below is a table showing the rank correlation coefficients among these 10 ratios belonging to four different classes of risk measurement. The upper triangular part of the matrix (values above the diagonal) represents Kendall's coefficient of rank correlation and the lower triangular part of the matrix represents Spearman's coefficient of rank correlation.

Table 1: Spearman's and Kendall's Rank Correlations

		Kendall's tau									
	Correlations	Sharpe	M2	Treynor	Jensen	Information Ratio	Sortino	Modified Sharpe Historical VaR	Modified Sharpe Normal VaR	Modified Sharpe CFVaR	Modified Sharpe Expected Shortfall
Spearman's rho	Sharpe	1	1.000**	.899**	.889**	.801**	.894**	.822**	.989**	.834**	.896**
	M2	1.000**	1	.899**	.889**	.801**	.894**	.822**	.989**	.834**	.896**
	Treynor	.958**	.958**	1	.928**	.737**	.849**	.781**	.896**	.798**	.849**
	Jensen	.974**	.974**	.985**	1	.778**	.873**	.812**	.883**	.787**	.875**
	Information Ratio	.938**	.938**	.878**	.921**	1	.783**	.754**	.795**	.698**	.789**
	Sortino	.976**	.976**	.948**	.969**	.929**	1	.836**	.889**	.754**	.951**
	Modified Sharpe Historical VaR	.950**	.950**	.910**	.940**	.909**	.961**	1	.820**	.704**	.852**
	Modified Sharpe Normal VaR	.999**	.999**	.958**	.970**	.934**	.975**	.949**	1	.836**	.891**
	Modified Sharpe CFVaR	.956**	.956**	.932**	.928**	.876**	.910**	.870**	.958**	1	.758**
	Modified Sharpe Expected Shortfall	.976**	.976**	.945**	.968**	.928**	.995**	.967**	.974**	.908**	1
** Correlation is significant at the 0.01 level (2-tailed).											

As we can observe, we have very high Spearman's and Kendall's correlation values for all pairs of risk adjusted performance measures irrespective of their method of risk measurement. This is an astonishing result as the measures belonging to different risk measurement category have a different orientation of risk and accordingly a different risk quantification technique. But these high correlations show that it does not matter much as to how an investor perceives or measures risk as long as ranking of the schemes is concerned. The lowest correlation that we observe is between Modified Sharpe Historical VaR and Modified Sharpe Cornish Fischer VaR as 0.704. The highest correlation that we observe is between Sharpe ratio and M^2 ratio at 1. All correlation are statistically significant at 1% level of significance. This indicates the high strength of statistical positive relationship among these risk adjusted measures.

Keeping in view these highly positive and significant rank correlation results among risk adjusted performance evaluation measures, we suggest that the regulatory authority for mutual funds in India i.e. SEBI should mandate reporting one risk adjusted measure each from the four different classes of risk measurement on at least half yearly basis. This move can ensure that investors receive appropriate and adequate risk adjusted performance information of the mutual fund schemes whatever be their orientation of risk measurement. This shall further lead to informed investment making decisions by the investors in the growing mutual fund industry.

6. Conclusion and Implication

An informed investment decision can only be made in the presence of adequate and accurate information. When the doors of the Indian mutual fund industry were opened to the private players in 1993, the objective was to expand the business by widening and deepening the market for asset management products. Considering the industry is still relatively young, its evolution and growth over the past two decades is impressive. For providing a transparent and secure environment to investors in the collective investment vehicle like mutual funds, the role of SEBI which is bestowed with the responsibility of regulating this dynamic sector becomes very crucial. The position of nodal association like AMFI also assumes greater prominence and responsibility. Today, there are 47 mutual fund houses operating a plethora of schemes to suit diverse investor objectives. We tried to evaluate the extent and gravity of reporting and disclosure norms relating to performance assessment of mutual funds schemes in India. We attempted to find out the

prevalence, if any, of regulatory disclosure requirements with regard to risk adjusted performance of mutual funds in India. We did this by carrying out an examination of the websites of SEBI and AMFI for all relevant guidelines and regulations in this regard. We also referred to third party reports of industry bodies and consultancies for the same. Finally we aimed to identify the appropriate risk adjusted performance measures that should be made compulsory for reporting purposes on a periodic basis. We did so by analyzing the rank correlation coefficients among ten risk adjusted performance measures based on four different interpretation of the concept of risk. We conclude that there is a deficiency of reporting and disclosure requirements in India with regard to performance assessment of mutual fund schemes in India in general and risk adjusted performance assessment in particular. In fact, we could not find out any mention of risk adjusted performance assessment in the existing reporting and disclosure norms. On the basis of our analysis of rank correlation among the ten risk adjusted measures spread across four different orientations of risk, we found a very high degree of positive and significant correlation among all measures irrespective of their orientation of risk measurement. Hence, we suggest that periodic reporting and disclosure of at least one measure from each of the four categories of risk measurement be made mandatory by the regulatory authority. This can serve the purpose of risk adjusted performance assessment for all investors irrespective of their orientation of risk measurement. We also suggest that the regulatory reporting and disclosure requirements of performance assessment be made more stringent and clear for mutual fund schemes in India. This can go a long way in ensuring that investors make informed investment decisions and the markets become more competitive. This study highlights the importance of accurate and precise disclosure requirements for assessment of financial performance of a mutual fund scheme.

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Appendix

Table 2: Descriptive Statistics

<u>Funds</u>	<u>Mean</u>	<u>Median</u>	<u>Maximum</u>	<u>Minimum</u>	<u>Std. Dev.</u>	<u>Skewness</u>	<u>Kurtosis</u>	<u>Jarque-Bera</u>	<u>Probability</u>	<u>Sum</u>	<u>Sum Sq. Dev.</u>	<u>Observations</u>
Birla SL Buy India Fund(G)	0.013753	0.024031	0.247859	-0.26637	0.068349	-0.75741	5.816621	50.71411	0	1.636591	0.551252	119
Birla SL Dividend Yield Plus(G)	0.012292	0.01924	0.274498	-0.23676	0.066658	-0.43138	6.19891	54.4295	0	1.462742	0.524308	119
Birla SL Equity Fund(G)	0.01302	0.018455	0.29537	-0.24616	0.075002	-0.38139	5.276984	28.59213	0.000001	1.549336	0.663792	119
Birla SL Frontline Equity Fund(G)	0.014879	0.018982	0.274923	-0.23653	0.066282	-0.4268	6.017929	48.77279	0	1.770545	0.518408	119
Birla SL Index Fund(G)	0.010384	0.010865	0.243357	-0.30844	0.072278	-0.75547	6.246707	63.58575	0	1.235652	0.61644	119
Birla SL India Opportunities Fund(G)	0.010473	0.019424	0.25443	-0.27257	0.070285	-0.76603	6.087772	58.91262	0	1.246316	0.58292	119
Birla SL Midcap Fund(G)	0.014654	0.025506	0.36374	-0.30509	0.080166	-0.26261	7.079401	83.88198	0	1.743801	0.758331	119
Birla SL MNC Fund(G)	0.016742	0.02456	0.167969	-0.2403	0.061477	-0.9087	5.271968	41.97136	0	1.992327	0.445971	119
Birla SL New Millennium Fund(G)	0.009317	0.015476	0.196903	-0.25833	0.072045	-0.56744	4.384066	15.88439	0.000355	1.108699	0.61248	119
Birla SL Tax Savings Fund(G)	0.00948	0.020196	0.300817	-0.37627	0.079959	-1.17496	8.504037	177.5904	0	1.128078	0.754426	119
DSPBR India T.I.G.E.R Fund-Reg(G)	0.012133	0.011485	0.283486	-0.28836	0.08219	-0.36957	4.724105	17.44775	0.000163	1.443793	0.797116	119
DSPBR Opportunities Fund-Reg(G)	0.013224	0.020717	0.233181	-0.24917	0.067592	-0.71544	5.278968	35.9037	0	1.573691	0.539097	119
DSPBR Technology.com Fund-Reg(G)	0.012199	0.017969	0.197061	-0.26386	0.073935	-0.83446	5.37523	41.78395	0	1.451693	0.645042	119
DSPBR Top 100 Equity Fund-Reg(G)	0.013722	0.012445	0.215715	-0.21742	0.06476	-0.48985	4.79337	20.70588	0.000032	1.63295	0.49487	119
Franklin India Bluechip Fund(G)	0.013298	0.018754	0.22922	-0.24358	0.065087	-0.69346	5.608279	43.26984	0	1.582463	0.499886	119
Franklin India Flexi	0.014185	0.015794	0.265971	-0.2597	0.069692	-0.5410	5.722913	42.56732	0	1.688019	0.573123	119

Cap Fund(G)				2		1						
Franklin India Index Fund-NSE Nifty(G)	0.010789	0.01352	0.246321	-0.30639	0.071666	-0.73653	6.285005	64.26585	0	1.283844	0.606046	119
Franklin India Opportunities Fund(G)	0.011751	0.014512	0.255709	-0.25531	0.074826	-0.653	5.077327	29.85377	0	1.398371	0.66067	119
Franklin India Prima Fund(G)	0.013278	0.026701	0.311808	-0.30669	0.077234	-0.70312	6.779363	80.62797	0	1.580121	0.703889	119
Franklin India Prima Plus Fund(G)	0.01505	0.023269	0.256542	-0.24762	0.066287	-0.60912	5.903315	49.15375	0	1.790928	0.51849	119
Franklin India Taxshield(G)	0.014003	0.020332	0.229986	-0.22769	0.064738	-0.64308	5.482463	38.75839	0	1.666327	0.494536	119
Franklin Infotech Fund(G)	0.011128	0.021182	0.18484	-0.22638	0.073056	-0.5221	3.705479	7.874176	0.019505	1.324273	0.629779	119
HDFC Capital Builder Fund(G)	0.013105	0.018529	0.243203	-0.28635	0.069227	-0.84869	6.170078	64.11373	0	1.559453	0.565494	119
HDFC Core & Satellite Fund(G)	0.011467	0.020784	0.299825	-0.33141	0.077494	-0.51875	6.588919	69.20213	0	1.364534	0.70862	119
HDFC Equity Fund(G)	0.01467	0.021057	0.290242	-0.27829	0.07328	-0.36697	5.692497	38.61646	0	1.745734	0.633661	119
HDFC Growth Fund(G)	0.013265	0.020826	0.255573	-0.25449	0.067646	-0.46109	5.400214	32.78176	0	1.57849	0.53997	119
HDFC Index Fund-Nifty(G)	0.009855	0.010199	0.24226	-0.28702	0.070473	-0.71282	5.967054	53.72788	0	1.172725	0.586035	119
HDFC Index Fund-Sensex Plus(G)	0.012381	0.013869	0.232478	-0.22779	0.065384	-0.38033	4.851405	19.8646	0.000049	1.473323	0.504454	119
HDFC Index Fund-Sensex(G)	0.01001	0.011602	0.237734	-0.28302	0.06953	-0.68511	5.796584	48.08772	0	1.191144	0.570465	119
HDFC Large Cap Fund-Reg(G)	0.009593	0.018339	0.255795	-0.27697	0.071289	-0.71217	5.851922	50.38762	0	1.141577	0.599695	119
HDFC Long Term Adv Fund(G)	0.011752	0.019454	0.252979	-0.2591	0.065898	-0.56446	5.885305	47.59733	0	1.398547	0.512414	119
HDFC Premier Multi-Cap Fund(G)	0.010897	0.016498	0.302989	-0.3358	0.075128	-0.60108	7.503087	107.7097	0	1.296746	0.666021	119
HDFC TaxSaver(G)	0.012498	0.014475	0.256595	-0.29163	0.07055	-0.59451	6.060012	53.43812	0	1.48722	0.587317	119
HDFC Top 200 Fund(G)	0.014396	0.018584	0.258652	-0.25175	0.06966	-0.34041	4.982393	21.78386	0.000019	1.71308	0.572605	119
ICICI Pru Dynamic Plan-	0.014287	0.016879	0.183617	-0.26764	0.064692	-0.87609	5.643165	49.86334	0	1.700168	0.493837	119

Reg(G)												
ICICI Pru FMCG Fund- Reg(G)	0.0157 01	0.0165 02	0.20458 7	- 0.2253 5	0.0598 01	- 0.5522 4	5.5546 48	38.407 69	0	1.8684 6	0.4219 81	119
ICICI Pru Index Fund- Reg(G)	0.0116 52	0.0127 92	0.24742	-0.3016	0.0715 45	- 0.7077 1	6.1729 97	59.853 51	0	1.3865 78	0.6040 05	119
ICICI Pru Long Term Savings Plan- Reg(G)	0.0104 61	0.0113 33	0.16348 8	- 0.1967 4	0.0470 75	- 0.8079	6.3086 19	67.223 83	0	1.2448 09	0.2614 93	119
ICICI Pru LT Equity Fund (Tax Saving)- Reg(G)	0.0123 79	0.0165 83	0.26880 7	- 0.2865 6	0.0758 92	- 0.5774	5.2125 24	30.884 54	0	1.4731 12	0.6796 3	119
ICICI Pru Midcap Fund- Reg(G)	0.0127 68	0.0209 89	0.32477 6	- 0.4332 5	0.0885 73	- 1.0285 8	8.4054 16	165.85 82	0	1.5193 74	0.9257 29	119
ICICI Pru Multicap Fund- Reg(G)	0.0126 68	0.0181 66	0.21427 9	- 0.2815 7	0.0711 64	- 0.9161 6	5.4596 63	46.644 87	0	1.5075 46	0.5975 92	119
ICICI Pru Technology Fund- Reg(G)	0.0130 73	0.0207 26	0.19033 2	- 0.3251 8	0.0786 42	- 0.8606	5.1434 36	37.469 34	0	1.5556 85	0.7297 8	119
ICICI Pru Top 100 Fund- Reg(G)	0.0125 06	0.0110 15	0.19260 3	- 0.2507 7	0.0646 09	- 0.7482 2	5.2754 1	36.774 92	0	1.4882 62	0.4925 73	119
ICICI Pru Value Discovery Fund- Reg(G)	0.0160 49	0.0235 38	0.28681	- 0.2985 3	0.0772 16	- 0.4775 5	5.8923 43	46.002 79	0	1.9097 85	0.7035 6	119
ICICI Pru Very Aggressive- Reg(G)	0.0104 19	0.0085 6	0.20005 2	- 0.2557 5	0.0591 94	- 0.9252 5	6.7868 74	88.083 67	0	1.2399	0.4134 67	119
LIC Nomura MF Equity Fund(G)	0.0085 33	0.0108 26	0.27952 5	- 0.3066 9	0.0799 79	- 0.6692 4	5.9640 28	52.444 35	0	1.0154 48	0.7548 07	119
LIC Nomura MF Growth Fund(G)	0.0101 42	0.0107 24	0.31048 3	- 0.3030 4	0.0774 09	- 0.5003 4	6.8357 09	77.915 39	0	1.2068 84	0.7070 8	119
LIC Nomura MF Index Fund- Nifty Plan(G)	0.0085 98	0.0088 45	0.23597 3	- 0.3060 6	0.0703 85	- 0.7444 1	6.3696 4	67.289 89	0	1.0232 07	0.5845 78	119
LIC Nomura MF Index Fund- Sensex plan(G)	0.0092 76	0.0101 56	0.23633 3	- 0.2728 3	0.0700 81	- 0.8248 7	6.2311 28	65.260 84	0	1.1038 96	0.5795 36	119
LIC Nomura MF Tax Plan(G)	0.0083 52	0.0142 6	0.27046 7	-0.2747	0.0735 63	- 0.7323	5.9660 53	54.256 65	0	0.9938 57	0.6385 59	119
Principal Dividend Yield	0.0085 5	0.0118 14	0.23851 1	- 0.2645 4	0.0708 95	- 0.8096 2	5.9398 47	55.853 83	0	1.0174 17	0.5930 87	119

Fund(G)												
Principal Growth Fund(G)	0.009479	0.014167	0.200727	-0.30083	0.073402	-0.99173	5.550024	51.74861	0	1.127989	0.635763	119
Principal Index Fund-Nifty(G)	0.010211	0.012796	0.243084	-0.30805	0.071321	-0.75651	6.334467	66.48096	0	1.215061	0.600224	119
Principal Tax Saving Fund	0.010974	0.014135	0.196626	-0.31864	0.077344	-1.0236	5.625789	54.96717	0	1.305964	0.705882	119
Reliance Banking Fund(G)	0.014765	0.014039	0.342422	-0.22216	0.087802	0.120373	4.212652	7.578725	0.02261	1.757093	0.909694	119
Reliance Diver Power Sector Fund(G)	0.011901	0.014293	0.29094	-0.26477	0.085811	0.187884	4.301274	9.096139	0.010588	1.416245	0.868902	119
Reliance Equity Opportunities Fund(G)	0.015403	0.028287	0.2816	-0.27007	0.075184	-0.57448	5.378245	34.59018	0	1.833013	0.667011	119
Reliance Growth Fund(G)	0.013983	0.018914	0.292382	-0.25277	0.075599	-0.25995	4.811642	17.61373	0.00015	1.663987	0.674395	119
Reliance Media & Entertainment Fund(G)	0.010192	0.014484	0.277003	-0.36066	0.082855	-0.71406	6.161492	59.67147	0	1.212891	0.810069	119
Reliance NRI Equity Fund(G)	0.013521	0.016448	0.291157	-0.24716	0.070883	-0.34799	5.839383	42.37635	0	1.608987	0.592885	119
Reliance Pharma Fund(G)	0.01862	0.020725	0.231432	-0.29404	0.072785	-0.80585	6.242265	65.00313	0	2.215795	0.625128	119
Reliance Reg Savings Fund-Equity Plan(G)	0.013887	0.013862	0.323631	-0.27385	0.079355	0.023092	5.591983	33.32252	0	1.652585	0.743075	119
Reliance Vision Fund(G)	0.012384	0.010869	0.274429	-0.22676	0.07472	-0.30625	4.401731	11.6025	0.003024	1.473729	0.658812	119
Sundaram Growth Fund(G)	0.009753	0.016782	0.265016	-0.26347	0.077117	-0.52567	5.039699	26.109	0.000002	1.160619	0.701755	119
Sundaram S.M.I.L.E Fund(G)	0.01443	0.026152	0.345326	-0.32222	0.087849	-0.32134	5.566049	34.69664	0	1.717172	0.910667	119
Sundaram Select Focus(G)	0.011609	0.015441	0.285141	-0.23771	0.073065	-0.02968	5.312801	26.53985	0.000002	1.381503	0.629942	119
Sundaram Select Midcap(G)	0.016754	0.029385	0.416489	-0.27239	0.082545	0.149664	7.94006	121.4484	0	1.993771	0.804012	119
Tata Dividend Yield Fund(G)	0.012755	0.016981	0.218423	-0.25406	0.068231	-1.06374	6.321326	77.13854	0	1.517893	0.549351	119
Tata Equity Opportunities Fund(G)	0.012404	0.015431	0.299917	-0.30449	0.075767	-0.60923	6.565389	70.39169	0	1.47611	0.677402	119
Tata Equity P/E Fund(G)	0.014059	0.017378	0.266982	-0.2926	0.076469	-0.58765	5.219757	31.28045	0	1.673006	0.690008	119
Tata Ethical Fund(G)	0.013564	0.018732	0.27157	-0.3505	0.075656	-1.1610	8.236635	162.7038	0	1.614083	0.675411	119

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Tata Index Fund-Nifty Plan(G)	0.010903	0.01246	0.244431	-0.30888	0.071927	-0.73995	6.256033	63.42636	0	1.297447	0.610476	119
Tata Index Fund-Sensex Plan(G)	0.010196	0.012371	0.245659	-0.27222	0.069134	-0.57821	5.619422	40.65169	0	1.213349	0.563977	119
Tata Infrastructure Fund(G)	0.010725	0.009573	0.303644	-0.30379	0.081951	-0.20633	5.090245	22.50794	0.000013	1.276316	0.792484	119
Tata Mid Cap Growth Fund(G)	0.01289	0.022554	0.321847	-0.30173	0.075475	-0.45437	6.716278	72.57285	0	1.533922	0.672182	119
Tata Pure Equity Fund(G)	0.012933	0.013828	0.265419	-0.2493	0.066081	-0.56852	6.351619	62.10909	0	1.538985	0.515268	119
Taurus Bonanza Fund(G)	0.008812	0.014649	0.339265	-0.35614	0.079634	-0.56568	7.912639	126.0111	0	1.048625	0.748312	119
Taurus Discovery Fund(G)	0.00709	0.023712	0.365736	-0.4361	0.096997	-0.85129	7.29357	105.7788	0	0.843704	1.110203	119
Taurus Star Share Fund(G)	0.012189	0.016959	0.39948	-0.35794	0.087768	-0.2648	7.520832	102.7287	0	1.450529	0.908974	119
Taurus Tax Shield Fund(G)	0.00934	0.011954	0.331915	-0.26642	0.081319	-0.17331	5.520864	32.10471	0	1.111464	0.780307	119
T-Bill Yield	0.005605	0.005883	0.008711	0.00253	0.001361	-0.72188	2.813443	10.50798	0.005227	0.66694	0.000219	119
BSE-500 Returns	0.010502	0.011662	0.287626	-0.31616	0.077016	-0.6664	6.322024	63.52727	0	1.249723	0.69991	119